# DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

# HERIOT-WATT UNIVERSITY

## F78AA1 Actuarial and Financial Mathematics A

**Date: 18 December 2013**

**Time: 13.00 to 15.00**

| Question | Marks |
|---|---|
| 1 | 12 |
| 2 | 9 |
| 3 | 5 |
| 4 | 14 |
| 5 | 8 |
| 6 | 12 |
| Total Marks | 60 |

**Attempt ALL SIX questions.**

**Approved electronic calculators may be used.**

This version created on: November 18, 2013

**1.** (a) Prove from first principles that:

$$(Ia)^{i}_{\overline{n}|} = \frac{\ddot{a}^{i}_{\overline{n}|} - nv^{n}}{i}.$$

[4 marks]

(b) Evaluate the following at interest of 9% *per annum* effective:

(i) The accumulation at time $t = 20$ years of an annuity of £1,000 *per annum*, payable annually in arrears for 20 years.

(ii) The present value now of a deferred annuity of £500 *per annum*, payable half-yearly in arrears for 10 years and deferred for 5 years.

(iii) The present value now of an annuity with term 10 years, commencing at £1,000 *per annum* and increasing by £100 at the end of each year, payable quarterly in arrears (so the first eight payments are £250, £250, £250, £250, £275, £275, £275, £275 and so on).

[8 marks]

**[Total 12 marks]**

**2.** You borrow £200,000 to buy a house. The loan is to be repaid over 30 years by level annual repayments, payable in arrears. The interest rate is initially 2% *per annum* effective, for the first three years, rising to 5% *per annum* effective thereafter. The interest rates can be changed by the lender at any time.

(a) (i) Show that the level annual repayment is £11,988.73. (Minor discrepancies due to rounding will not be penalised.)

(ii) Show that the capital outstanding just after the 10th repayment has been made is £149,406.08. (Minor discrepancies due to rounding will not be penalised.)

[5 marks]

(b) Just after the 10th repayment has been made, you lose your job and decide to retrain by studying for a Masters degree. You ask the lender if you can miss the next two payments while you return to university. The lender agrees, but will charge interest at 8% *per annum* effective during this two-year period. Calculate the new level annual repayment, when they recommence.

[4 marks]

**[Total 9 marks]**

[PLEASE TURN OVER]

**3.** (a) Given an equation of value $f(i) = 0$ in respect of a financial contract, what conditions must a solution $i^*$ of this equation satisfy to be defined as the yield of the financial contract?

[3 marks]

(b) State a general condition on the cashflows under a financial contract such that a yield will always exist.

[2 marks]

**[Total 5 marks]**

**4.** A government issues a bond on 1 April 2013, with term 20 years and coupon 8% *per annum* payable half-yearly in arrears. The bond will be redeemed at par.

(a) The market prices the bond at issue to give a gross redemption yield of 10% *per annum* effective. Show that the issue price per £100 nominal is £84.63.

[4 marks]

(b) An investor subject to income tax at 20% and capital gains tax at 30% buys some of the bonds on the issue date. What net redemption yield *per annum* effective does she obtain? [6 marks]

(c) Just after the 20th coupon has been paid, she sells the bonds to another investor, at a price such that the net yield she obtains on the completed transaction is 10% *per annum*. At what price per £100 nominal did she sell the bonds? [4 marks]

**[Total 14 marks]**

**5.** On 1 January 2009, an investment fund had assets worth £100,000. On 1 January 2010 it received an inflow of £7,000; on 1 January 2011 it paid out £4,000; on 1 January 2012 it received an inflow of £10,000, and on 1 January 2013 it had assets worth £120,000. Just before the money was paid out on 1 January 2011, the fund had assets worth £106,000.

(a) Calculate the linked internal rate of return over the calendar years 2009–2012, based on the subdivision of the period into two parts, one being the calendar years 2009–2010 and the other being the calendar years 2011–2012.

[6 marks]

(b) The fund had assets worth £105,000 on 1 January 2010, and assets worth £108,000 on 1 January 2012, in each case just before the inflows of cash on those dates. Calculate the time-weighted rate of return over the calendar years 2009–2012.

[2 marks]

**[Total 8 marks]**

[PLEASE TURN OVER]

**6.** An investor has a plan to develop an open-cast mine. Buying the land and preparing the ground will take two years, and has an initial, one-off cost of £10 million. The mine will yield profits of £4 million a year for five years, with the first profit cashflow being received exactly two years after the start of the project. Six years after the start of the project, the investor will incur costs of £2 million to clean up the site. The investor has to choose between two ways of raising the loan capital of £10 million.

(A) The investor could borrow the money from a bank. The bank would charge 7% *per annum* effective interest on capital outstanding and would pay interest at 5% *per annum* effective on any positive balances deposited with it.

(B) The investor could issue a bond, with term six years, and coupons of £1 million *per annum* payable annually in arrears, to be redeemed at par. At any time, the investor would be able to borrow additional money from the bank at 7% *per annum* effective interest and deposit money with the bank at 5% *per annum* effective interest.

By calculating the accumulated profit after six years under each method of financing the project, determine which method the investor should choose.

**[Total 12 marks]**

[END OF PAPER]